ADNI Alzheimer's Disease Neuroimaging Initiative

ADNI – Brain Health Registry
Computerized Testing and Longitudinal Monitoring

Weiner, Michael MD[1]; Nosheny, Rachel PhD[2]; Fockler, Juliet[1]
[1]Department of Radiology and Biomedical Imaging, University of California, San Francisco
[2]Department of Psychiatry, University of California, San Francisco

Contents

Summary

The Brain Health Registry (BHR, PI: M Weiner) is a University of California, San Francisco IRB approved online registry and database for recruiting, screening, and longitudinal monitoring of participants for clinical neuroscience studies. The overall goal of the BHR is to accelerate the development of new treatments for brain diseases by facilitating clinical research. BHR captures extensive health, cognitive, and lifestyle data using self-report and study partner-report questionnaires and assessments.

The BHR offers optional, online assessments to be completed unsupervised by ADNI participants in their homes. Additionally, participants have the ability to invite their study partners to the Caregiver and Study Partner Portal (CASSP) within BHR, where the study partner will answer questions about the participant and the study partner him/herself.

Participants are invited to register at home for the ADNI-BHR within two weeks of their in-clinic visit. Immediately following their registration, the participant will be asked to take an online brain test, answer questions, and invite their ADNI study partner to the BHR- Caregiver and Study Partner Portal. All of this should take about 15-20 minutes. Subsequently they will receive an email inviting them to return every three months to answer more questions and take repeated tests.

Overall Goals. The overall goal is to study the utility of remote, online data collection in a clinical sample of older adults.

Specific aims

Aim 1: To collect longitudinal data from ADNI participants using the BHR. Data will be collected using self-report online questionnaires and a self-administered online neuropsychological test (NPT). All ADNI - cognitively normal (CN) and mild cognitive impartment (MCI) - participants will be invited to join BHR, and follow-up data will be collected at 3-month intervals for the duration of the ADNI study.

Aim 2: To collect longitudinal data from study partners of ADNI participants, who may be caregivers, through a Caregiver and Study Partner (CASP) Portal within the BHR. Information collected will include data about the participant (recent changes in cognitive function and everyday functioning), data about the study partner (self-report questionnaire data and NPTs), and data about effects of caregiving on caregivers (caregiving burden and caregiver stress). Follow-up data will be collected at 6-month intervals for the duration of the ADNI study.

Aim 3: To validate online self- and CASP-reported data against data obtained in-clinic.

Aim 4: To share all data collected by BHR with the scientific community through USC/LONI/ADNI website.

Method

I. **ADNI-BHR participants**
   a. *Questionnaires.* ADNI-BHR study participants will complete BHR questionnaire modules including self-reported demographics, cognition, medical conditions, current medications, and various lifestyle factors. Questionnaires currently used are based on well-validated instruments, including The Everyday Cognition Scale (ECog) and the Cognitive Function Instrument (CFI).
   b. *Cognitive Test.* Memtrax MMT is a short (~ 3 min) N-Back test using complex visual stimuli. This continuous recognition test uses a slide-show in which 50 images are presented, of which 25 are repeated. Participants respond by recording whether an image was a repetition.

II. **ADNI-BHR study partners**
   a. *Questionnaires.* The CASPP includes 6 questionnaires, each of which takes 3-10 minutes to complete. The questionnaires can be broadly characterized as gathering information about the participant and about the study partner him/herself.
      i. Questions about the participant include a short health screener, and participant cognition based on well validated instruments, including the Everyday Cognition (ECog) scale; Cognitive Function Instrument (CFI), Informant version; and the Functional Assessment Questionnaire (FAQ).
      ii. Questions about the study partner him/herself include demographics, short health screener, stress, and study partner relationship to participant. Study partners who identify as caregivers also complete a caregiver burden module based on well-validated instruments.

All ADNI-BHR data is uploaded to the LONI website quarterly by participant PTID.

Dataset Information

This methods document applies to the following dataset(s) available from the ADNI repository:

Dataset Name	**Date Submitted**
BHR Data Dictionary	February 20, 2018
Brain Health Registry: Baseline Questionnaire	February 20, 2018
Brain Health Registry: Everyday Cognition	February 20, 2018
Brain Health Registry: Longitudinal Questionnaire	February 20, 2018

Brain Health Registry: Memtrax	February 20, 2018
Brain Health Registry: Study Partner ADL	February 20, 2018
Brain Health Registry: Study Partner Caregiver Burden	February 20, 2018
Brain Health Registry: Study Partner Everyday Cognition	February 20, 2018
Brain Health Registry: Study Partner FAQ	February 20, 2018
Brain Health Registry: Study Partner Initial	February 20, 2018
Brain Health Registry: Study Partner Relationship	February 20, 2018
Brain Health Registry: Study Partner Study Confirmation	February 20, 2018

About the Authors

This document was prepared by Dr. Michael Weiner, Dr. Rachel Nosheny and Juliet Fockler, University of California, San Francisco (UCSF). For more information please contact Juliet Fockler at 415-221-4810 ext. 24253 or by email at Juliet.fockler@ucsf.edu.

Notice: This document is presented by the author(s) as a service to ADNI data users. However, users should be aware that no formal review process has vetted this document and that ADNI cannot guarantee the accuracy or utility of this document.